

BISRAT SERUR

AI/ML Engineer | Data Engineering & Analytics | Full-Stack AI Product Development

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Portfolio: bistratasfaw.github.io/Portfolio-HTML-CSS-1 • LinkedIn: linkedin.com/in/bistratasfaw • GitHub: github.com/bistratasfaw

PROFESSIONAL SUMMARY

AI/ML engineer and full-stack builder who ships production systems end to end — from PyTorch model training and rigorous evaluation through FastAPI services, Celery data pipelines and production Next.js interfaces. Trained and deployed a Siamese U-Net satellite change-detection model with bootstrap-validated, leakage-audited metrics; engineered geospatial and RSS ingestion pipelines over Sentinel-2 imagery and 100+ live sources; shipped monetized LLM features on the Anthropic and OpenAI SDKs. Fluent in Python (PyTorch, FastAPI, Celery, Polars/DuckDB) and TypeScript (Next.js, React, Prisma, PostgreSQL/PostGIS), with production Stripe billing, OAuth/JWT auth, GDPR tooling and Sentry/Prometheus observability. MS in Computer Science in progress.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, C#, C, Bash

AI / Machine Learning: PyTorch, segmentation-models-pytorch, scikit-learn, Albumentations, NumPy, transfer learning, semantic segmentation, computer vision, Siamese/U-Net architectures, focal-Tversky & Dice loss, Hidden Markov Models (hmmlearn), model evaluation & threshold calibration, bootstrap confidence intervals, cross-validation, data-leakage auditing

Generative AI / LLM Engineering: Anthropic Claude SDK, OpenAI API (gpt-4o-mini), prompt engineering, LLM feature design & gating, AI agent skills and AGENTS.md operating specs (Claude Code), Leonardo AI image generation, AI-assisted development workflows

Data Engineering: ETL/ELT pipeline design, Celery + Celery Beat, Redis, Cloudflare Workers & Vercel Cron, scheduled ingestion, idempotent upserts, deduplication, reverse ETL, Polars, DuckDB, Parquet, pandas, Alembic & Prisma migrations, geospatial raster/vector processing (GDAL, rasterio, GeoPandas, Shapely), eodag / Copernicus Sentinel-2, RSS & HTML scraping, REST integrations (Shopify, Stripe, OANDA, Alpha Vantage, NewsAPI)

Data & Analytics: PostgreSQL, PostGIS, SQL Server, MySQL, SQLite, Supabase, Neon, Prisma, SQLAlchemy, schema design & normalization, time-series aggregation, read-through caching, KPI metrics, CTR & funnel analytics, Recharts, D3.js, Plotly, CesiumJS, quantitative scoring models, Sharpe/Sortino/Calmar, max drawdown, expectancy

Backend, Frontend & APIs: FastAPI, Node.js, Express, Next.js 15/16 (App Router) API Routes, REST, WebSockets, Server-Sent Events, Pydantic & Zod validation, JWT/OAuth 2.0, HMAC request signing, webhooks, rate limiting; React 18/19, TypeScript, Tailwind CSS, shadcn/ui, Radix, TanStack Query & Table, Zustand, Vite

Cloud, DevOps & Quality: Docker & Docker Compose, Vercel, Cloudflare (Pages, Workers, KV, Tunnel), AWS S3/boto3, Linux, Nginx, Gunicorn/Uvicorn, Git & GitHub, Sentry, Prometheus, Flower, Pytest, Vitest, Playwright, Locust load testing, Ruff, MyPy, ESLint

Security & Compliance: OAuth 2.0, JWT/JWKS, bcrypt, Fernet field encryption & key rotation, SSRF protection, CSP & security headers, Stripe webhook signature verification, GDPR Art. 17/20 tooling, WCAG 2.1 AA

SELECTED ENGINEERING PROJECTS

TerrAlert (TerrAlert.io) — Satellite Change-Detection SaaS (Computer Vision + Geospatial Data Platform) | *Python, PyTorch, FastAPI, Celery, PostGIS, Docker*

- Trained and shipped a **Siamese U-Net** change detector (shared ImageNet-pretrained ResNet-34 encoder) producing per-pixel change masks from co-registered 10 m Sentinel-2 image pairs, using PyTorch, segmentation-models-pytorch and Albumentations.
- Lifted model performance **~4.5x over baseline** with two-stage transfer learning (LEVIR-CD pretrain to OSCD fine-tune) and a focal-Tversky loss that fixed recall collapse — **F1 0.059 to 0.266**; current production checkpoint scores **F1 0.330 (95% CI 0.306–0.355), IoU 0.198** on a held-out 324-chip SpaceNet7 test set via 2,000-sample bootstrap.
- Established a leakage-proof evaluation protocol (threshold locked on validation, single-pass test, city-level held-out splits, bootstrap CIs) and documented two data-leakage post-mortems plus a negative result — a 2.3x chip expansion that **lowered** F1 to 0.245 with non-overlapping CIs — keeping inflated metrics out of production.
- Engineered the geospatial ETL: **eodag/Copernicus Sentinel-2 ingestion**, GDAL/rasterio/GeoPandas raster-vector processing, training-chip generation across three public benchmarks, PostGIS **ST_AsMVT** vector tiles, self-hosted **Tiler** tiling and corroboration feeds (NASA FIRMS, GLAD/RADD).
- Built the distributed backend on **FastAPI + Celery/Redis** with Celery Beat scan scheduling per subscription tier, Flower monitoring, Alembic migrations and a Docker Compose stack (PostgreSQL 15/PostGIS, Redis 7, Tiler, workers), gating inference on checkpoint presence so the pipeline degrades cleanly rather than failing.
- Hardened for production: Supabase **JWT/JWKS** verification, HMAC-verified **Stripe** billing across five tiers, Fernet field encryption with key rotation, SSRF allow-listing, **GDPR Art. 17/20** endpoints, **Sentry + Prometheus**, Locust load tests and verified nightly backups.

Vows & Volts (vowsandvolts.com) — Multi-Tenant Marketplace SaaS with AI Concierge | *TypeScript, Next.js 16, React 19, Prisma 7, PostgreSQL 16, Stripe Connect*

- Architected a four-role platform (couple, vendor, admin, support) spanning ~59 routes with **NextAuth v5** session revocation, **Stripe Connect + Tax + Billing** marketplace payments, Upstash Redis rate limiting, Vercel Blob storage, Resend/Twilio messaging and `next-intl` i18n.
- Integrated **Anthropic Claude** as an in-product AI concierge and **Leonardo AI** for generative venue re-imagination, and authored reusable Claude Code agent skills and `AGENTS.md` operating specs to standardize AI-assisted development across repositories.
- Built a **Playwright QA automation harness**: four role-scoped browser contexts x three viewports producing ~170 screenshots and 27 interaction scripts, using accessible-selector-first queries and a reload-and-re-query persistence assertion that exposes optimistic-UI false passes.

- Led a database migration after reproducing shadow-DB migration failures and prepared-statement corruption under concurrent queries causing intermittent 500s — moved to native PostgreSQL 16 with git history preserved, plus deploy-time migration and bootstrap pipelines.

Newsletter Intelligence Platform — LLM-Powered News Aggregation SaaS | *Next.js 15, TypeScript, Cloudflare Workers, Neon Postgres, OpenAI*

- Built a serverless ingestion pipeline pulling **100+ RSS sources across 26 categories** on a **5-minute Cloudflare Worker cron**, with URL-level deduplication, DOMPurify sanitization, per-category limits and 7-day retention — persisted to Neon serverless Postgres, cached via Cloudflare KV and CDN.
- Shipped **OpenAI gpt-4o-mini article summarization** as a monetized Pro-tier feature behind runtime feature flags, and a non-LLM personalization engine computing decay-weighted category-preference scores from reading history.
- Implemented passwordless email-OTP auth (Better-Auth), Stripe Checkout/webhooks/portal, build-time **Zod environment validation** via `instrumentation.ts` that blocks startup on misconfiguration, CSP nonces, HSTS, Sentry and Vitest unit/integration/E2E suites.
- Instrumented product and operational analytics: affiliate impression/click tracking with UTM tagging and CTR-by-placement reporting, plus API latency, DB query performance, RSS fetch and webhook success monitoring with threshold alerting.

PerAI View — Real-Time Geospatial AI-Usage Analytics Platform | *React, CesiumJS, D3, Node/TypeScript, FastAPI, scikit-learn, PostGIS, Redis*

- Designed a polyglot streaming architecture — an Express/TypeScript ingestion API that anonymizes users to city-level geography, a **FastAPI + scikit-learn classifier microservice** returning category and confidence scores, and **Redis pub/sub to WebSocket** fan-out on a 5-second cadence.
- Built the analytics layer over PostgreSQL/PostGIS (global, regional and time-series endpoints behind a read-through Redis cache with 1-hour TTL and write-triggered invalidation), visualized on a **CesiumJS 3D globe** with D3 dashboards; IPs are hashed and deleted within 24 hours.
- A heatmap and usage data of AI platform based on geolocation.

APEX Dropship AI — Multi-Tenant Shopify Commerce Automation SaaS | *Next.js, Supabase Postgres, Clerk, Stripe, Anthropic SDK, Vercel*

- Implemented a bi-directional **Shopify Admin REST (2025-01)** integration: OAuth install flow, a scheduled **2-hour Vercel Cron** sync pulling catalogs and 90-day order history with rate-limit pacing (2 req/s, 40 burst), idempotent upserts on composite unique keys, and reverse-ETL write-back of price and inventory.
- Secured the OAuth handshake with **stateless HMAC-SHA256 state tokens** (10-minute expiry) that remove server-side session storage while staying unforgeable, and modeled a per-tenant sync state machine capturing status, last-sync time and error detail.

InvAdvisor — Equity Screening & Multi-Factor Scoring Workstation | *Next.js 15, React 19, TypeScript (strict), Prisma, SQLite, TanStack*

- Designed five weighted quantitative scoring models (Future-Proof, Financial Health, Growth Quality, Valuation, Risk) over a normalized 7-table schema of quarterly/annual fundamentals, valuation snapshots and ownership data, with scores precomputed by batch jobs and denormalized to keep screening interactive across a 50,000-ticker design universe.
- Built the analytics UI (TanStack Query/Table, Recharts, `decimal.js` precision) over Zod-validated paginated REST endpoints with saved multi-factor filter presets and side-by-side comparison; tested with Vitest and React Testing Library.

Quantitative Research & Algorithmic Trading Systems | *Python (Polars, DuckDB, Numba, hmmlearn), FastAPI, Node.js, OANDA v20*

- Architected a deterministic, local-first backtesting platform: a three-tier immutable data lake (raw to clean to features) on **Parquet + DuckDB + Polars**, Numba-accelerated compute, portable SQLite `opt` experiment archives, mandatory walk-forward and Monte Carlo validation, **HMM market-regime classification**, and regression tests asserting bit-identical reproducibility behind an 80% coverage gate.
- Built a live EUR/USD execution service (Node/Express) with four pluggable market-data backends (OANDA v20, Alpha Vantage, Yahoo, simulator), EMA(9/21) + RSI(14) signals, fixed-fractional 0.5% risk sizing and a 3% daily drawdown breaker — paper by default behind a two-step arm-code and preflight interlock before any live order.
- Formalized a discretionary multi-timeframe (4H/1H to 5M to 1M) gold strategy into quantified rules — liquidity sweeps, break-of-structure, FVG/inverse-FVG detection, Fibonacci and equilibrium zones, SMT divergence vs DXY/XAG — with an SSE-streamed dashboard and CSV/OANDA replay backtester scored on win rate, drawdown, expectancy and R-multiple.

PROFESSIONAL EXPERIENCE

Founder & Lead Engineer — Independent AI & Data Platform Development | *Silver Spring, MD | Jan 2025 – Present*

- Own the full lifecycle of the platforms above — problem framing, data acquisition, model training and evaluation, pipeline and backend engineering, front-end delivery, billing, security hardening and production operations.
- Operate as a one-person engineering org with senior discipline: migration-managed schemas, typed and validated boundaries, automated test and QA harnesses, structured observability, documented architectural decisions and post-mortems.

Operations & Analytics Lead — Prime Store Trading PLC | *Jun 2025 – Dec 2025*

- Directed operations and process-optimization initiatives, introducing data-driven decision making and performance reporting across the business.

Operations & Systems Manager — Job Trading PLC | *Aug 2023 – Dec 2025*

- Managed operational systems and analytics initiatives, implementing scalable workflows and performance-monitoring practices.

Founder & Operations Director — Bisrat Asfaw Imports | *Jun 2021 – Dec 2025*

- Coordinated international logistics, supplier relationships and inventory management; improved reliability through process redesign and analytical problem solving.

EDUCATION

Master of Science, Computer Science — University of the Potomac | *In Progress*

Bachelor of Science, Computer Science — HiCoE School of Computer Science and Technology | *Addis Ababa, Ethiopia*

Additional: AI-directed engineering — agent skills, operating specs and automated verification harnesses. Chinese (intermediate).